

FINAL EXAM REQUIREMENTS

The final exam is conducted in the form of a **project defense focused on Machine Learning System Design**.

1. Introduction

The final exam is conducted in the form of a project defense. Each student (or group) presents their project according to schedule, followed by a short Q&A session.

Schedule - I will upload soon.

Students must submit three final deliverables:

1. Presentation (according to schedule and time limit)
2. Written Report (PDF)
3. GitHub Repository with Source Code

2. Evaluation Criteria

2.1 Actuality & Relevance

1. Clearly describe the problem the project aims to solve.
2. Explain the importance and real-world significance of the chosen topic.
3. Explain potential impact (business / social / technical / scientific/medicine)

2.2 Novelty & Originality

1. Describe what is new or innovative in your approach.
2. Show how your work differs from standard examples or known solutions.

2.3 Related Work

1. Provide an overview of existing research or analytical approaches.
2. Compare your solution with previous studies or known methods (if it is possible).

2.4 Methods, Algorithms, Metrics, and Datasets (METHODOLOGY)

Data & ML Components

- Data sources and data collection
- Data preprocessing & feature engineering
- Selected ML algorithms and justification
- Evaluation metrics and reasoning

System Design & Architecture (MANDATORY)

- Overall **system architecture diagram** (end-to-end pipeline)
- Data flow: input → processing → model → output
- Description of how the system would work in production

System Logic (Front Perspective)

- Describe how the system works from a **user/business point of view**
- Show how model predictions are translated into **decisions (e.g., approve/reject, risk level)**
- Clarify how the results are **used in practice and what value they create**

2.5 Results

1. Present the outcomes of your analysis or models.
2. Explain the results clearly and interpret them.
3. Discuss limitations and potential improvements.

2.6 Visualization

1. Provide meaningful graphs, charts, and tables.
2. Visualizations must be clear, readable, and relevant to the conclusions.

2.7 Presentation Quality

Evaluation includes not only content but also **delivery quality**:

- Logical and structured storytelling
- Clear explanation of technical concepts

- Quality of slides (readability, design, clarity)
- Time management
- Ability to answer questions

2.8 Presentation

Your presentation must include:

1. Problem statement
2. Actuality and Relevance
3. Novelty and originality
4. Literature overview (related work)
5. Methodology
6. Results
7. Visualizations
8. Conclusions

2.9 Written Report

Must include all sections above

Length: 8–10 pages (PDF)

Must include **architecture diagrams and system description**

2.10 Source Code (GitHub Repository Required)

All students must submit their source code via GitHub.

Code Requirements:

1. Clean, documented, reproducible code
2. Notebooks must run from start to finish

3. Final

Each student must submit at Teams

1. Presentation (PDF/PPTX)
2. Final Report (PDF)
3. GitHub Repository Link - (show at 2nd page of report)